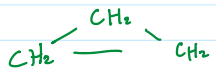
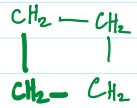
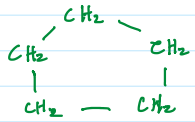
February 16, 2026 5:12 PM

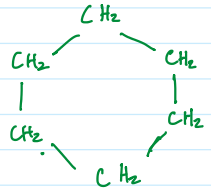
4) Cyclic Hydrocarbon

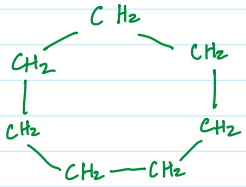
↳ A hydrocarbon whose molecules have closed ring structures

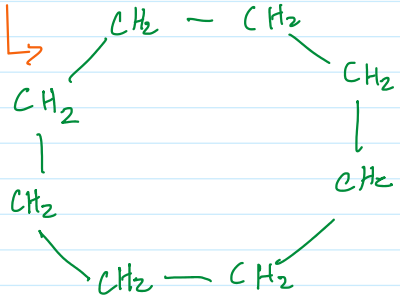
↳  Cyclopropane

↳  Cyclobutane

↳ 

↳  Cyclohexane

↳  Cyclo heptane

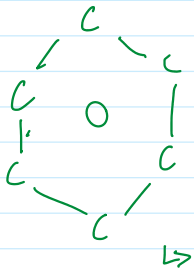
↳  Cyclo-octane

Naming:

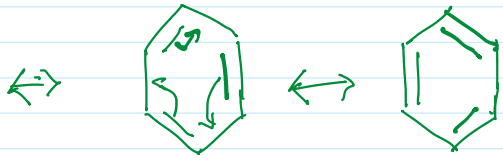
1) Find the parent chain
2) Use the cyclo-prefix
3) When numbering, a double or triple bond takes priority over the branches
4) For naming branches, use alphabetical order.

5) Aromatic Hydrocarbons.

↳ A compound with a structure based on benzene ring: a ring of 6 carbon atoms.



Benzene
C_6H_6



[Resonance "alternating double bond"]

↳ The benzene ring is considered to be the parent molecule with alkyl group named as branches attatched to the benzene.

c

↳ The benzene ring is considered to be the parent molecule, with alkyl group named as branches attatched to the benzene

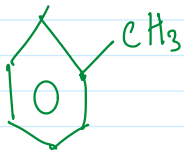
↳ When 2 or more groups are attatched to the benzene **ring**, the **lowest** possible number is to be given to any branch.

↳ for some aromatic molecules where the attatched group is not easily named, it is more convinient to consider the benzene ring as a branch rather than the parent molecule.

↳ when the benzene ring is the attatched branch, C_6H_5 is called the phenyl **group**

Benzene = Phenyl (benzene)

↓



parent with branches

$CH_3 - CH - CH_2 - CH_2 - CH_3$

↳ when numbering parent chain, the **double and triple bonds take priority.**